

Name: _____

Date: _____

TRIGONOMETRY, (EXACT TRIGONOMETRIC VALUES)

LO: I can recall and use exact values of sin, cos, and tan for 0, 30, 45, 60, and 90 degrees.

Exact trigonometric values

The **exact values** of sin, cos, and tan at key angles (0° , 30° , 45° , 60° , 90°) can be expressed as **surds or fractions** without a calculator.

Key facts: $\sin 30^\circ = \frac{1}{2}$, $\cos 60^\circ = \frac{1}{2}$, $\tan 45^\circ = 1$. These come from equilateral triangles and isosceles right-angled triangles.

Quick examples

●	$\sin 30^\circ = \frac{1}{2}$	half the hypotenuse
●	$\cos 45^\circ = \frac{\sqrt{2}}{2}$	from isosceles right triangle
●	$\tan 60^\circ = \sqrt{3}$	from equilateral triangle
●	$\sin 45^\circ = \frac{\sqrt{2}}{2}$	$\sin 45^\circ = \frac{\sqrt{2}}{2}$
●	$\cos 30^\circ = \frac{\sqrt{3}}{2}$	$\cos 30^\circ = \frac{\sqrt{3}}{2}$

Key idea

Learn the exact values from the 30-60-90 triangle (sides 1, $\sqrt{3}$, 2) and the 45-45-90 triangle (sides 1, 1, $\sqrt{2}$).

$$\sin 60^\circ = \frac{\sqrt{3}}{2}, \cos 60^\circ = \frac{1}{2}$$

Derived from half an equilateral triangle

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - finding sin values

State the exact value of $\sin 60^\circ$.

Use the 30-60-90 triangle: sides 1, $\sqrt{3}$, 2

$$\begin{aligned} \sin 60^\circ &= \text{opposite/hypotenuse} = \\ &= \frac{\sqrt{3}}{2} \end{aligned}$$

$$\frac{\sqrt{3}}{2}$$

The side opposite 60° is $\sqrt{3}$, hypotenuse is 2.

Example 2 - finding cos values

State the exact value of $\cos 45^\circ$.

Use the 45-45-90 triangle: sides 1, 1, $\sqrt{2}$

$$\begin{aligned} \cos 45^\circ &= \text{adjacent/hypotenuse} = 1/ \\ &= \frac{\sqrt{2}}{2} \text{ (rationalised)} \end{aligned}$$

$$\frac{\sqrt{2}}{2}$$

Both sin and cos of 45° are equal.

Example 3 - finding tan values

State the exact value of $\tan 30^\circ$.

$$\begin{aligned} \tan &= \sin/\cos = (\frac{1}{2}) / (\frac{\sqrt{3}}{2}) \\ &= \frac{1}{\sqrt{3}} \\ &= \frac{\sqrt{3}}{3} \text{ (rationalised)} \end{aligned}$$

$$\frac{\sqrt{3}}{3}$$

$$\tan = \text{opposite/adjacent} = 1/\sqrt{3}$$

Example 4 - using exact values

Find the exact length of the side opposite 30° in a triangle with hypotenuse 10.

$$\begin{aligned} \sin 30^\circ &= \text{opposite} / \text{hypotenuse} \\ \frac{1}{2} &= \text{opposite} / 10 \\ \text{opposite} &= 5 \end{aligned}$$

$$5$$

Use the exact value of $\sin 30^\circ = \frac{1}{2}$.

YOUR TURN – PRACTICE (deliberate practice)

1. Recall sin values

State the exact value of each.

- a) $\sin 0^\circ$
- b) $\sin 30^\circ$
- c) $\sin 45^\circ$
- d) $\sin 90^\circ$

2. Recall cos values

State the exact value of each.

- a) $\cos 0^\circ$
- b) $\cos 30^\circ$
- c) $\cos 60^\circ$
- d) $\cos 90^\circ$

3. Recall tan values

State the exact value of each.

- a) $\tan 0^\circ$
- b) $\tan 30^\circ$
- c) $\tan 45^\circ$
- d) $\tan 60^\circ$

4. Using exact values in triangles

Find the exact length.

- a) Hypotenuse 8, angle 30° . Find opposite side.
- b) Hypotenuse 6, angle 60° . Find adjacent side.
- c) Adjacent 5, angle 45° . Find opposite side.
- d) Opposite 4, angle 30° . Find hypotenuse.

5. Mixed exact values

Evaluate without a calculator.

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) $\sin 30^\circ = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

b) $\sin 45^\circ = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

c) $\tan 45^\circ = 1$ ☐

Correct answer: _____

Reason: _____

d) $\cos 30^\circ = \frac{1}{2}$ ☐

Correct answer: _____

Reason: _____

e) $\sin^2\theta + \cos^2\theta = 1$ for all angles ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) An equilateral triangle has sides of **10 cm**. Using exact values, find its height and area. Leave your answer in surd form.

Expression: _____

Simplified: _____

b) Without a calculator, show that $\sin^2 60^\circ + \cos^2 60^\circ = 1$. Then find the exact value of $\sin 60^\circ \times \cos 30^\circ + \sin 30^\circ \times \cos 60^\circ$.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

